Urban design, transport, and health 2

Land use, transport, and population health: estimating the health benefits of compact cities

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Using a health impact assessment framework, we estimated the population health effects arising from alternative land-use and transport policy initiatives in six cities. Land-use changes were modelled to reflect a compact city in which land-use density and diversity were increased and distances to public transport were reduced to produce low motorised mobility, namely a modal shift from private motor vehicles to walking, cycling, and public transport. The modelled compact city scenario resulted in health gains for all cities (for diabetes, cardiovascular disease, and respiratory disease) with overall health gains of 420–826 disability-adjusted life-years (DALYs) per 100 000 population. However, for moderate to highly motorised cities, such as Melbourne, London, and Boston, the compact city scenario predicted a small increase in road trauma for cyclists and pedestrians (health loss of between 34 and 41 DALYs per 100 000 population). The findings suggest that government policies need to actively pursue land-use elements—particularly a focus towards compact cities—that support a modal shift away from private motor vehicles towards walking, cycling, and low-emission public transport. At the same time, these policies need to ensure the provision of safe walking and cycling infrastructure. The findings highlight the opportunities for policy makers to positively influence the overall health of city populations.

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This is the second in a Series of three papers about urban design, transport, and health

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Introduction

Cities around the world are dealing with the consequences of changing population demographics and policies that have failed to effectively manage the associations between land use, mobility, and population health. Urban growth (the expansion of metropolitan areas) and the pressure it places on urban infrastructure is now a major international challenge. By 2050, the populations of Australia's four largest cities are predicted to be similar to Australia's current total population,[1] and the USA, China, and India will see predicted increases in the populations of their large cities of 33%, 38%, and 96%, respectively.[2]

Continued population growth is associated with ever-increasing demands on transport systems. Governments increasingly emphasise the need to integrate transport and land-use planning,[3] acknowledging that land-use decisions substantially influence transport options and travel choices. Sprawling residential-only developments that dominate most suburban areas in North America, Australia, and New Zealand limit the ability of people to walk or cycle for their daily travel requirements.[4] In these countries, low-density housing developments make public transport development cost prohibitive, producing a reliance on private motorised transport and increasing exposure to the risks associated with traffic speed, traffic volume, vehicle emissions, and physical inactivity.[5] In response to economic growth, private car use is also dramatically increasing in many middle-income countries such as Brazil,[6] China, and India.[7] Resultant declines in physical activity and increases in air pollution, noise, and risk of motor vehicle crashes combine to produce increased rates of chronic disease and injury.[8]

City planners and policy makers—who have the power to influence the health of rapidly expanding cities and increasingly motorised populations—need to prioritise the minimisation of health risk exposures while maintaining or enhancing the mobility of city residents. Recent innovations in transportation have generated an expectation of a transportation revolution, in which road deaths, serious injury, and congestion are eliminated because of web connectivity, automated vehicles, and advanced software. Like an engineering fix for global warming, this vision is seductive and will eventually play a part in the resolution of current transportation challenges. However, serious obstacles including software viruses, security risks, and the need for fall-back options in the case of major connected system failures mean that technological solutions will be achieved, but only over the coming decades. Additionally, these

Key messages

- Considerable health gains are observed by city planning that encourages a compact city—namely, a city of short distances that promotes increased residential density, mixed land use, proximate and enhanced public transport, and an urban form that encourages cycling and walking.
- A compact city approach also results in reduced city-specific particulate emissions owing to reduced motor-vehicle emissions.
- Policies that incentivise walking, cycling, and public transport while reducing subsidies for private motor vehicle use will influence the health and sustainability of growing cities.

solutions will not address the broader health and environmental consequences associated with land use, the transport system, and rapid motorisation identified in the first paper in this Series,[9] namely increased cardiovascular disease, diabetes,[10] and respiratory disease[11,12] coupled with escalating road traffic injury[13] and the ongoing challenges associated with infectious diseases in highly urbanised areas.[14]

Deaths from road traffic injury increased worldwide by 46% from 1992 to 2012, making it the eighth leading cause of death.[15] The UN General Assembly resolution on global road safety acknowledges the emerging challenges associated with reducing road injury,[16] as do initiatives such as Sustainable Safety and Vision Zero. These efforts are consistent with the UN Post-2015 Sustainable Development Agenda, which emphasises the risks associated with global trends towards urbanisation, and disaster risk reduction and mitigation.

For more on **Sustainable Safety** see https://www.sustainablesafety.nl

For more on **Vision Zero** see https://www.visionzeroinitiative.com

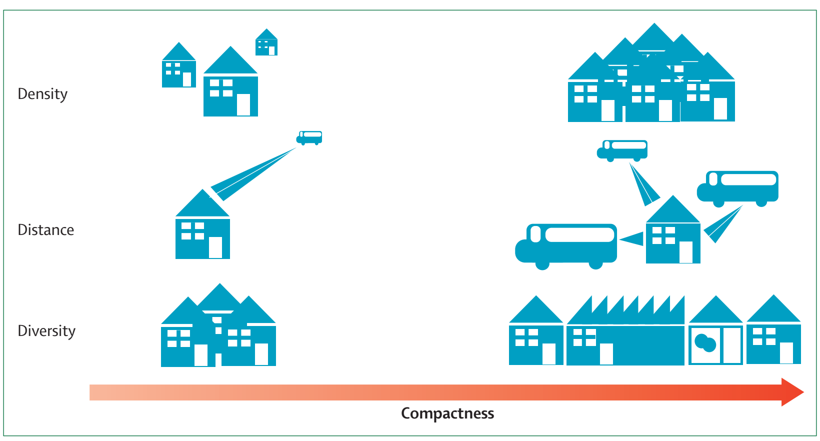

Figure 1: **Illustration of the terms density, distance, and diversity as applied in the compact cities model**

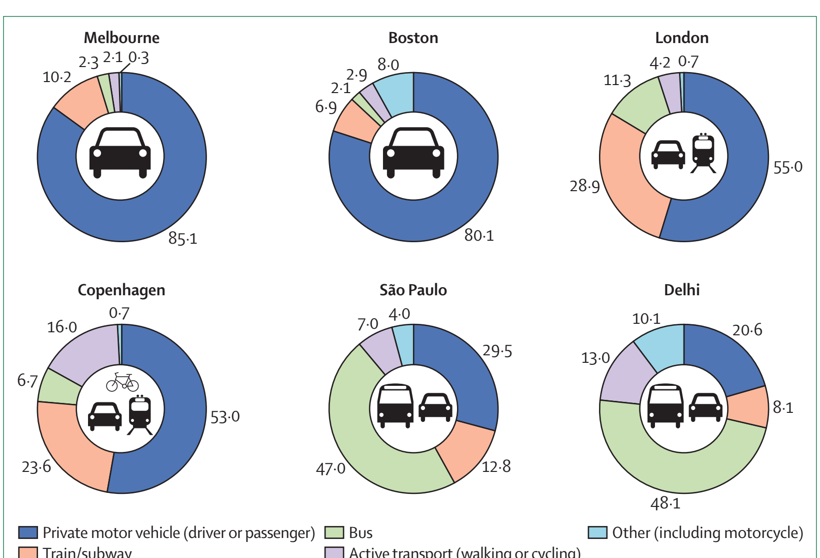

Figure 2: **Percentage of vehicle kilometres travelled (VKT) by mode in each city at baseline with dominant transport modes depicted**

However, these efforts have rarely acknowledged the impact of land-use issues (such as urban growth) on travel mode choice or travel distance, and the consequent effects on population and environmental health.

The effects of land use and transport mode choice on population health are not well described, partly because they occur against a backdrop of complex, interacting, and dynamic environmental, technological, and population conditions that evolve over years. Building on elements of the first paper in this Series,[9] this paper investigates the population health outcomes associated with land-use policies that influence a city's transport mode choice. We model urban design interventions that aim to create a compact city and quantify the potential health gains that residents of compact cities would obtain through adoption of low motorised mobility.

The compact cities model

Key features and characteristics

We assessed the associations between land use, transport, and population health by selecting the key elements presented in the preceding paper in this Series (see the pathway diagram in Giles-Corti and colleagues)[9] for which information and relationships have been demonstrated.[17] We also reviewed the literature to identify measures of association for these key elements. Having obtained estimates of the association between land use and transport mode choice, we applied a health impact assessment framework[18] to produce a model for which estimates of population health outcomes were derived.

We used characteristics from six cities to model the city-specific effect of land use and urban design interventions on transport mode choice and population health. We selected the cities based on a combination of the country's stage of development, level of motorisation, geographical disparity, and the availability of reliable transport and health data. The cities selected were Melbourne, Australia (high-income and highly motorised); Boston, MA, USA (high-income and moderately motorised); London, UK (high-income and moderately motorised); Copenhagen, Denmark (high-income and moderately motorised); São Paulo, Brazil (upper-middle-income and moderately motorised); and Delhi, India (lower-middle-income and rapidly motorising).[19] We applied weighted average associations between urban design (density, distance, and diversity) and transport mode choice for each city that were derived from a previous meta-analysis.[17] The associations ranged from 0·02 to 0·29 per unit change in the relationship between density, distance, and land-use diversity and the respective transport mode choice. Density refers to population density or residential unit; distance refers to the average distance to public transport; and diversity refers to the land-use mix within a given precinct (eg, mix of commercial and residential land use) (figure 1). Estimates derived from the meta-analytic approach included studies with small samples that were predominantly from North American cities, and included uncontrolled confounding factors as a result of study design and limited control groups. Therefore, an element of uncertainty is associated with these estimates.[17]

For each city, we assessed the influence of land use, urban design interventions, and transport mode choices on population health outcomes—namely road trauma (road deaths and serious injury; ICD-AM [International Classification of Diseases, Australian modification] V00–V89), cardiovascular disease (ICD-AM I00–I99), diabetes (ICD-AM E10–E14), and respiratory disease (ICD-AM J30–J98). For comparative purposes, road trauma and chronic disease health outcomes (cardiovascular disease, diabetes, and respiratory disease) were reported as disability-adjusted life-years (DALYs), which are a combination of the sum of the years of potential life lost due to premature mortality and years of productive life lost due to a disability.[20]

The key determinants of population health associated with transport mode choice as identified in the literature and applied within the model were risk of death or injury per kilometre travelled by mode;[21–23] level of physical activity (measured by metabolic equivalents [METs][24,25]) expended by mode choice per h[26–29] and its effect on cardiovascular disease and type 2 diabetes; and chronic exposure (via inhalation) to fine particulate matter (PM) associated with exhaust emissions and raised dust from transport.[30,31] Baseline population, transport mode share, road trauma, levels of physical inactivity, and air quality data were input for each city from the most recent data sources available.

Modelled changes in road trauma

Travel-mode, road deaths, and serious injury data were available from recent travel surveys and government agencies in Melbourne,[32,33] Delhi,[34] São Paulo,[35] London,[36] Boston,[37] and Copenhagen.[38,39] These data included distance travelled per mode and per day, and road fatalities and injuries by mode per year (injury data for Delhi were adjusted to account for historical under-reporting[40]). To estimate differences in road trauma as a result of change in mode share, we applied an approach that combined the probability of a crash between different road users given the changing proportions of transport modes within the transport system.[21,22,41,42] We based estimates of chronic disease burden associated with road trauma on country-level data from the Global Burden of Disease Study 2013[43] converted to city-specific DALY estimates scaled to city population size.

Modelled changes in chronic disease

We modelled levels of transport-related physical inactivity and vehicle emissions for each city, and estimated changes in physical activity by calculating changes in the average estimated distance travelled per person, per mode, and per day. We then applied consistent average METs per h associated with each form of travel for each city.[24] We calculated average time spent in each travel mode using a combination of average speed by mode and daily distance travelled for each city.

The effect of increased physical activity (in METs per week) on cardiovascular disease and diabetes was estimated using linear associations established in the literature of 0·25 and 0·20 per 1000 kcal per week, respectively,[44–49] with a benefit threshold restricted to activity in excess of 2·5 METs per h.[50] We considered this method to be appropriate for the small variations in overall physical activity that we modelled. However, a meta-analysis[51] of studies that incorporated greater proportional changes than ours showed associations between non-vigorous physical activity and disease risk to be non-linear. In addition, the modelling of physical activity did not take into account demographic differences (eg, age profiles and sex) in the likelihood of a modal shift between passive and active transport modes, the effects of exercise compensation,[52,53] or differences in average speeds or METs associated with different demographic groups. Although these differences have been explored elsewhere,[54] we considered the levels of uncertainty associated with the estimates to be too great to be reliable at anything other than population levels.

We estimated total vehicle emissions in each city by obtaining the most recently available city-specific particulate emissions (diameter <10 μm [PM_{10}] and diameter <2·5 μm [$PM_{2·5}$]) data and estimates of the proportion of particulate matter generated by motor

	Melbourne		São Paulo		Delhi		London		Boston		Copenhagen	
	Deaths	Injuries	Deaths	Injuries	Deaths	Injuries	Deaths	Injuries	Deaths	Injuries	Deaths	Injuries
Vehicle driver	0·2	7·3	1·7	38·1	0·4	2·5	0·2	3·5	0·9	2·2	0·3	3·7
Vehicle passenger	0·2	7·1	1·9	106·7	0·4	2·5	0·2	3·5	0·5	1·7	0·3	3·5
Train, tram, or subway passenger	0·1	0·2	0·0	0·1	1·5	8·7	0·0	0·2	0·0	0·1	0·1	0·6
Bus passenger	0·1	0·7	0·0	7·1	0·2	1·4	0·1	2·5	0·0	0·2	0·3	0·7
Walking	7·6	108·6	16·6	216·6	20·9	125·3	5·9	64·8	2·7	12·0	3·2	50·0
Cycling	1·4	79·8	25·8	472·7	4·3	25·8	4·4	140·8	2·5	23·0	0·6	26·6
Other (including motorcycle)	16·5	495·1	23·6	826·5	9·1	54·3	13·1	229·0	0·1	3·5	3·4	171·0

Data are per 100 million km.

***Table 1*: Risk of road death and injury per 100 million kilometres travelled by transport mode**

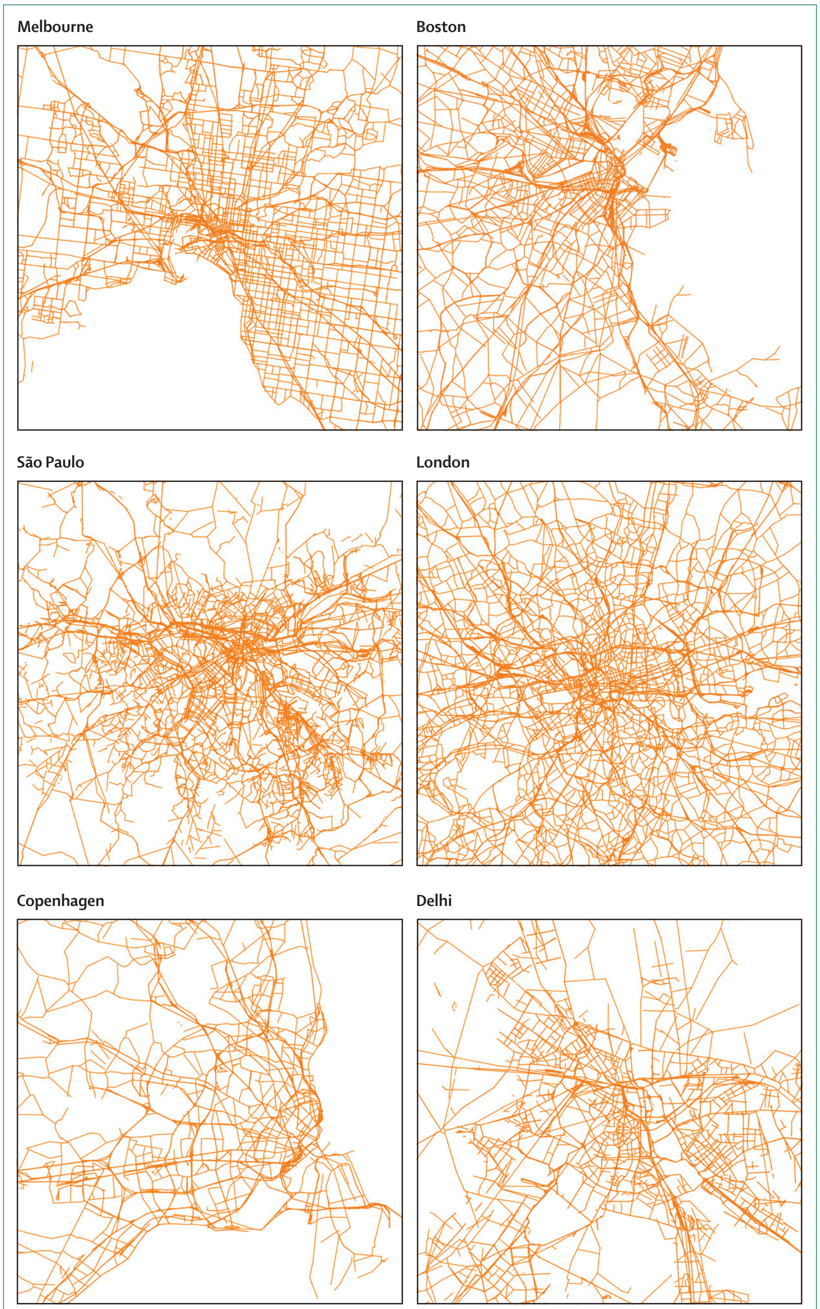

Figure 3: **Variation in city plans**
Melbourne and Boston* (approximately 4·5 million residents each) have grid-like road networks designed around the predominant use of motorised transport. London (8 million) and São Paulo (12 million) have medium levels of motorisation combined with dense spaghetti-like road networks. Copenhagen (0·6 million) and Delhi (17 million) have sparse road networks reflecting low levels of infrastructure dedicated to motorised transport. *Map scale twice as large as other cities.

See Online for appendix

vehicles through combustion and suspended road dust in Melbourne, Delhi, São Paulo, and London.[55–63] Data for the proportion of PM_{10} produced by vehicles for Boston and Copenhagen were unavailable so an estimate of 30% was used, which approximated the median of the other cities. To reflect the clean-fuel-technology bus fleets implemented in Delhi[63] and the urban bus renewal programme in São Paulo,[64] additional vehicle kilometres travelled (VKT) by bus were assumed to be undertaken in fleets powered by compressed natural gas, which emits negligible additional fine particle emissions.[65]

We modelled DALYs attributable to respiratory disease and cardiovascular disease associated with coarse ($<PM_{10}$) and fine ($<PM_{2·5}$) particulate emissions. Particulate emissions associated with vehicles were assumed to change proportionately with kilometres travelled. We estimated the effect of $PM_{2·5}$ reduction on long-term cardiovascular disease risk and the effect of PM_{10} reduction on respiratory disease risk using associations gathered from a systematic search of the literature. The search identified an approximate 20% increase in cardiovascular disease mortality risk per 10 μg/m³ increase in $PM_{2·5}$[66–70] and an approximate 2·5% increase in respiratory disease risk per 10 μg/m³ increase in PM_{10}[71–76] (although estimates for this effect vary widely between studies). We assumed that air pollution affected mortality and incidence to the same degree,[77] and that levels of PM_{10} and $PM_{2·5}$ were closely correlated[78,79] and their effects on cardiovascular disease were not additive. Particulate emissions from private motorised transport modes per person and per kilometre travelled[80,81] were assumed to be equivalent.

Correction for error

Application of this model involved coordination and linkage of multiple studies and effects across a wide variety of disciplines, each with their own inherent sources of error. We acknowledge these uncertainties and applied Monte Carlo simulation using Analytica 4.4 software (Lumina Decision Systems; Los Gatos, CA, USA) to each of the city-specific models. The simulation provided confidence interval boundaries for the estimated DALYs associated with population health outcomes, and sensitivity analyses to identify factors that were most influential in contribution to chronic disease and road trauma outcomes (appendix). To reflect the uncertainty associated with both land-use changes on transport mode choice and transport mode choice on the risk of road death or injury, we allowed the estimates associated with these variables to vary according to normal distributions with standard deviations equal to 20% of the mean.[82] For example, the effect of land-use density change on VKT was allowed to vary according to a normal distribution, as was the effect of VKT change on the risk of exposure to death or injury per kilometre travelled. Other variables we allowed to vary due to acknowledged real-world uncertainty[71] and probably skewed distributions were the proportion of active transport transferred from vehicles distributed between cycling and walking, total kilometres travelled per year (negatively skewed log-normal distribution),[50] METs associated with transport modes,[83] average speed by mode, and total particulate matter. The relationship between levels of physical activity and chronic disease and between exposure to particulate

	Melbourne		São Paulo		Delhi		London		Boston		Copenhagen	
	DALYs lost per 100 000 population	Total city DALYs	DALYs lost per 100 000 population	Total city DALYs	DALYs lost per 100 000 population	Total city DALYs	DALYs lost per 100 000 population	Total city DALYs	DALYs lost per 100 000 population	Total city DALYs	DALYs lost per 100 000 population	Total city DALYs
Cardiovascular disease	3277	136 622	4961	558 286	13 770	2 306 920	4579	374 251	5092	236 310	4315	24 261
Type 2 diabetes	606	25 265	1116	125 589	2996	501 927	368	30 077	868	40 282	976	5488
Respiratory disease	1642	68 457	1623	182 644	3927	657 900	2191	179 075	2126	98 663	2268	12 752
Road trauma	536	22 346	1447	162 838	2892	484 504	411	33 592	635	29 469	454	2553

Table 2: **Disability-adjusted life-years (DALYs) lost related to cardiovascular disease, diabetes, respiratory disease, and road trauma**

matter and chronic disease reflected a normal distribution. Nonetheless, the breadth of findings for many effects of land use on transport mode choice, or the effects of transport mode choice on the key outcomes (road trauma and chronic disease) are likely to vary beyond the conservative estimates made here. The model and underlying assumptions are available in the appendix.

City comparisons

We estimated the burden of disease and injury (with respect to road trauma, cardiovascular disease, diabetes, and respiratory disease) associated with land use and transport for each city, and calculated the proportion of VKT by travel mode (figure 2) and the risk of road death and serious injury per VKT (table 1). The proportion of VKT using private motor vehicles compared with public transport, cycling, and walking varied considerably between cities. Differences in VKT by travel mode partly reflect cities with substantial population size but low population densities, which have typically invested in road infrastructure and have high levels of private motorised transport (eg, Melbourne and Boston). Despite the contrast in transport mode choice within and between cities (figure 3), the considerable between-city differences in estimated risk of death or injury for similar transport modes should also be noted. For example, the estimated per kilometre risk of death of a driver in a private motorised vehicle in Delhi is twice the risk in Melbourne or London. Similarly, the estimated risk of death as a cyclist in São Paulo is 43 times greater than is the risk in Copenhagen (table 1).

DALYs associated with chronic disease (cardiovascular disease, diabetes, and respiratory disease) and road trauma (road deaths and serious injury) vary widely between cities (table 2), reflecting observations in the 2013 Global Burden of Disease Study.[43] For example, differences in DALYs for cardiovascular disease, diabetes, and road trauma are four to five times greater in the low-income to middle-income city of Delhi than in the high-income city of Melbourne. Estimated differences in respiratory disease between cities demonstrate smaller disparities, but the estimates for Delhi are almost twice those of other cities assessed. The disparity observed in cardiovascular disease between Delhi and the

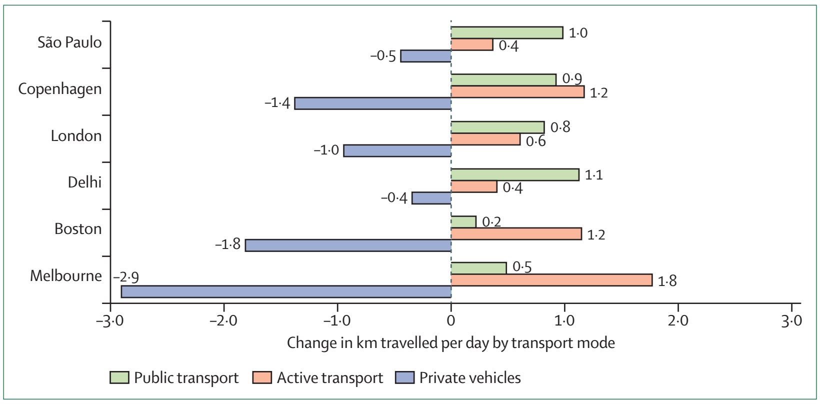

Figure 4: **Estimated change in total kilometres travelled per day by mode of transport under the compact cities model for each city**

remaining cities could reflect the presence of risk factors that contribute to cardiovascular disease beyond those associated with the transport system—such as socioeconomic constraints, diet, exercise, tobacco use, and, potentially, a genetic predisposition to cardiovascular disease later in life.[84]

Opportunities to enhance population health

We used our compact cities model[9] to model a scenario designed to enhance population health in the six selected cities. In what we refer to as the compact cities model—a city of short distances that promotes higher residential density, mixed land use, proximate and enhanced public transport, and an urban form that encourages cycling and walking[85]—we provided an alternative to each city's current land-use configuration and estimated the overall differences in the burden of disease under a compact city scenario. Under the compact cities model we increased land-use density by 30%, reduced average distance to public transport options by 30%, and increased diversity of land use by 30%. We combined these changes with an additional transport policy initiative that supported a 10% modal shift away from private motor vehicle driver and passenger VKTs (excluding motorcycles) to either cycling (two thirds of the total shift) or walking (a third of the total shift). This modal shift is similar to the goals of transport

	Melbourne	São Paulo	Delhi	London	Boston	Copenhagen
Change in travel-related METs per week (95% CI)	72·1% (38·9 to 119·5)	24·1% (–4·3 to 65·2)	18·5% (–6·7 to 54·4)	39·1% (10·4 to 78·6)	55·7% (26·8 to 99·0)	28·9% (–2·2 to 69·5)
Change in transport-related particulate emissions (95% CI)	–12·4% (–17·3 to –6·8)	–4·9% (–6·8 to –2·8)	–3·2% (–4·9 to –1·4)	–10·1% (–14·3 to –5·4)	–11·8% (–16·3% to –6·9%)	–10·9% (–15·3 to –5·8)

Other vehicles, including powered two-wheelers and three-wheelers, were not modelled within the scenarios because the proportion of travel within this mode was assumed to remain stable. All transport mode changes refer to change in vehicle kilometres travelled from baseline. METs=metabolic equivalents.

Table 3: **Changes in physical activity and particulate emissions associated with the compact cities model application**

	Melbourne	São Paulo	Delhi	London	Boston	Copenhagen
Cardiovascular disease (ICD-AM I00–I99)	622 (312 to 1071)	363 (14 to 915)	565 (169 to 1117)	582 (244 to 1053)	765 (355 to 1386)	337 (4 to 832)
Type 2 diabetes (ICD-AM E10–E14)	86 (40 to 159)	55 (–9 to 155)	28 (–10 to 91)	27 (7 to 61)	94 (41 to 189)	53 (–4 to 146)
Respiratory disease (ICD-AM J30–J98)	2 (1 to 4)	3 (1 to 5)	22 (8 to 42)	8 (4 to 14)	3 (–1 to 5)	2 (1 to 4)
Road trauma (ICD-AM V00–V89)	–34 (–64 to –7)	–4 (–71 to 62)	2 (–48 to 51)	–41 (–64 to –19)	–34 (–66 to –1)	–1 (–22 to 20)
Total	679 (330 to 1181)	420 (12 to 1029)	620 (167 to 1233)	581 (216 to 1084)	826 (352 to 1553)	393 (5 to 967)

Data are 50th percentile estimates (95% CI). Aggregated individual estimates may not equal the total due to rounding and Monte Carlo estimation. ICD-AM=International Classification of Diseases, Australian modification.

Table 4: **Disability-adjusted life-years (DALYs) gained per 100 000 population under the compact cities model**

policies currently being implemented in several European cities, such as Zurich, Switzerland, that impose barriers to private motor vehicle use.[86] The percentages of land-use changes and transport modal shifts across each city were selected on the basis that they were pragmatic and could be implemented over a reasonable timeframe. For example, in a city such as Melbourne, a 30% reduction in average distances to public transport—which could be achieved through prioritisation of transit-oriented development—means reduction of a journey from an average of 2 km to 1·6 km. That said, a 30% increase in land-use density in a city such as Delhi, which already has an estimated population density approaching 20 000 people per square kilometre,[87] is unrealistic (and probably unnecessary) compared with the same percentage increase in Melbourne or Copenhagen.

Given the land-use changes imposed under the compact cities model and the mode shift from private motor vehicles, increases observed in each city in public transport travel (trains, trams, and buses) and active transport (walking and cycling) are not surprising (figure 4). The largest changes in walking and cycling are seen in Melbourne, Boston, and Copenhagen. However, much of the increase in walking and cycling in these cities comes from building upon very low existing VKT in these modes. Therefore, the overall proportion of kilometres travelled by walking and cycling remains low (<10%), even after land-use changes are applied. Nonetheless, as a consequence of the modal shift to walking and cycling, increases in estimated travel-related physical activity (as measured by METs per week) are observed in these cities (table 3).

Changes in transport-related particulate emissions due to modal shifts away from private vehicles and towards low-emission public transport were observed for all cities (table 3). Although estimated emissions reduced most notably in the highly motorised cities of Melbourne (–12·4%), Boston (–11·8%), London (–10·1%), and Copenhagen (–10·9%), emissions also reduced (to a lesser extent) in São Paulo (–4·9%) and Delhi (–3·2%), where VKT transfer from private to public transport was proportionately lower.

Health gains were observed for all cities for cardiovascular disease, respiratory disease, and diabetes under the compact cities model (table 4). In addition to reductions in estimated emissions, these health gains were associated with enhancements to land use that brought about a modal shift towards walking and cycling, resulting in increased transport-related physical activity for all cities. The greatest gains in transport-related physical activity were in the highly motorised cities of Melbourne (72·1%) and Boston (55·7%), in which baseline active transport levels were low. Transport-related physical activity increases were also observed, albeit to a lesser extent, in London (39·1%), Copenhagen (28·9%), São Paulo (24·1%), and Delhi (18·5%).

Road trauma associated with serious deaths and injuries (DALYs per 100 000 people) was estimated to increase in Melbourne, Boston, and London under the compact city scenario (table 4). Differences in the other cities were estimated to be marginal or non-significantly reduced (table 4, figure 5). Increases in road trauma DALYs were a direct consequence of the modal shift to walking and cycling, which carry a higher per kilometre risk of death or injury than do private vehicles, even with the application of a safety in numbers[88] effect estimate. Although safety in numbers might reduce per kilometre risk, it should not be relied upon as a strategy to reduce absolute levels of road trauma.[89,90]

These road trauma estimates are limited to trauma associated with motor vehicle crashes. With the exception of São Paulo, the analysis does not include an estimate of the risk associated with cycle-only incidents. The inclusion of cycle-only crashes would probably add to the burden of road trauma associated with increases in cycle and pedestrian deaths and injuries.[91] Road deaths and serious injury can change as a function of long-term economic growth, which can lead to increased use of private motor vehicles (figure 6), or short-term economic cycles.[92] Consequently, elements of a city's economy could attenuate (or amplify) the population health outcomes. We have not adjusted the estimates of road trauma for the potential economic fluctuations in each of the cities.

Although modal shift toward active transport options resulted in increased road trauma, a considerable proportion of DALYs gained across the chronic diseases in the compact cities model were contributed to by policies that encouraged walking and cycling uptake rather than land-use changes alone. This effect was particularly evident in the highly motorised cities of Melbourne, Boston, and London, and underscores the importance of provision of additional transport policy, pricing, or regulatory incentives[93] to encourage active transport if reductions in chronic disease are to be realised. In cities with existing high levels of walking and cycling, such as Copenhagen and Delhi, most benefit was gained from land-use changes that produced increased rates of walking and cycling rather than the promotion of public transport alone.

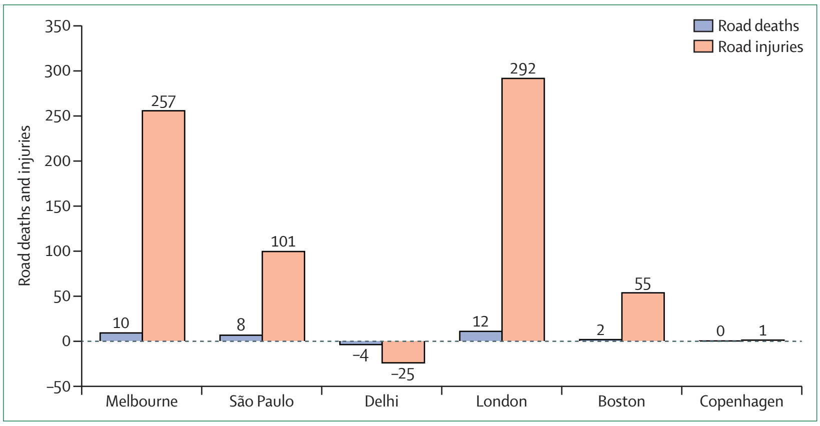

Figure 5: **Estimated change in road deaths and recorded injuries per year under the compact cities model for each city**

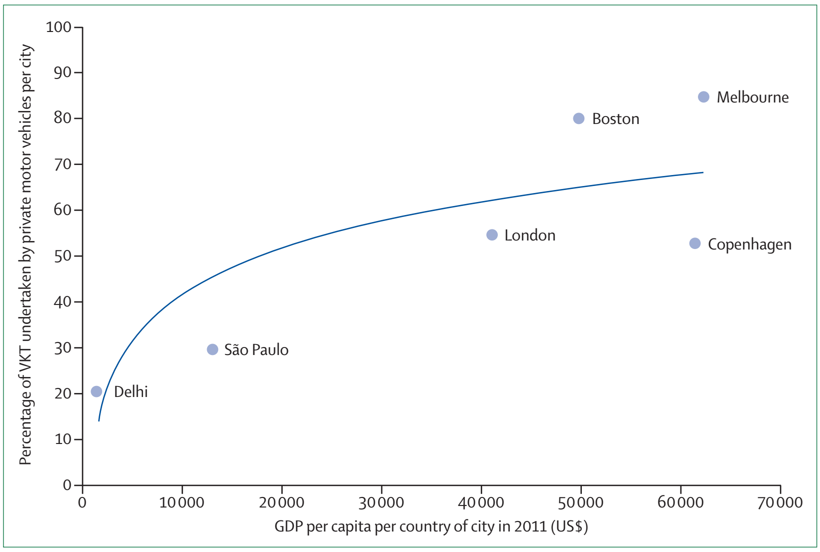

Figure 6: **Relationship between proportion of private motorised transport and gross domestic product (GDP) per person**

For cities with a high GDP (eg, London and Copenhagen), uptake of private motorised transport is only offset by investment in public transport and active transport infrastructure. To create a future transport system that enhances both mobility and health, cities such as Delhi and São Paulo have the opportunity to invest in public and active transport to reduce reliance on private motorised transport and avoid the mobility and negative health consequences associated with highly motorised cities. VKT=vehicle kilometres travelled.

Urban design and population health

Effect on lifestyle-related chronic disease

Many countries concerned by costs associated with the mounting burden of lifestyle-related chronic disease[94] have developed plans and public policy initiatives that encourage increased levels of physical activity.[95–97] Although the extent to which these plans are successfully disseminated, enacted, and monitored varies,[98–102] the findings we report here suggest that government policies need to actively pursue land-use planning and urban design interventions that encourage a modal shift toward walking, cycling, and low-emissions public transport to influence the overall health of growing city populations. From rapidly motorising to highly motorised cities, enhancements to urban design (such as increased land-use density, increased diversity, and a decrease in average distance to low-emission public transport) along with a modal shift of motorised trips to active transport options (such as walking and cycling) are essential to limit the rising burden of chronic disease associated with transportation systems. However, a move towards a compact city to mitigate the growing burden of chronic disease needs to be explored in more detail in cities such as Delhi and São Paulo where population density per square kilometre is already high. For example, these cities might respond better to other interventions such as access to proximate public transport or a mix of local destinations. In this paper, we have limited the modelling to a small number of urban and transport planning and design interventions, and we have limited the outcome to chronic disease. Infectious disease and heat-related mortality and morbidity are also direct consequences of excessive urban densities,[14,103] and other interventions

Extent of separation between motor vehicles and active transport users (pedestrians and cyclists)

(%)	0	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100
Melbourne	34	31	27	23	19	15	11	6	2	–3	–8	–14	–20	–26	–33	–41	–49	–59	–71	–88	–141
Boston	42	37	31	25	18	12	5	–2	–10	–18	–26	–35	–44	–55	–66	–78	–92	–108	–128	–155	–240
Delhi	–2	–15	–29	–43	–57	–73	–88	–105	–122	–140	–159	–180	–201	–225	–250	–279	–310	–347	–393	–455	–653
London	41	36	30	25	19	14	8	1	–5	–12	–20	–27	–36	–45	–54	–65	–77	–91	–109	–133	–208
Copenhagen	1	–4	–10	–16	–23	–29	–36	–43	–51	–59	–67	–76	–85	–96	–107	–119	–133	–149	–168	–196	–281
São Paulo	5	–3	–11	–19	–28	–37	–47	–57	–67	–78	–89	–101	–114	–128	–143	–160	–179	–201	–228	–266	–384

DALYs attributed to road trauma for each city

Figure 7: **Estimated effect of additional separation of active transport vehicle kilometres travelled (VKT) from traffic required to offset additional road trauma disability-adjusted life-years (DALYs) for each city under the compact cities scenario (positive numbers represent estimated increases in road trauma)**

(eg, destination accessibility and demand management) will be important.

Infrastructure investment for vulnerable road users

Despite estimated health gains in chronic disease associated with the compact cities model, the modal shift towards less private motorised transport and increased walking and cycling resulted in small to moderate increases in road trauma for Melbourne, London, and Boston. We estimated the extent of separated cycling and walking VKT needed to offset the estimated increase in DALYs associated with road deaths and serious injuries under the compact city scenario (figure 7).

The highly motorised cities of Melbourne, Boston, and London would need the largest investments in separated pedestrian and cycling infrastructure (equivalent to approximately 40%, 30%, and 35% of total pedestrian and cycling VKT, respectively) to offset the likely increase in road deaths and serious injuries associated with the compact city scenario (figure 5). By contrast, Delhi, Copenhagen, and São Paulo are estimated to require minimal additional infrastructure to see no net increase in injury burden. These between-city differences reflect the comparatively high proportion of existing walking and cycling in Copenhagen (16%), Delhi (13%), and São Paulo (7%) and the comparatively low level of increased risk associated with a high number of vulnerable road users; this low level is partly due to the large proportion of VKT also undertaken using public transport in Copenhagen (56%), Delhi (31%), and São Paulo (60%). Nonetheless, changes to infrastructure in these cities could still lead to further reductions in road trauma, and a growing number of vulnerable road users could increase pedestrian and cyclist deaths. However, increased injury among pedestrians and cyclists could be matched by injury reductions for other transport modes (eg, drivers and passengers).

The compact cities model presented in this paper shows a macro-level observation of the relationships between a limited number of land-use and urban design interventions and transport mode choices. The model provides limited insight with respect to the interactions between individuals in entire city populations and how these interactions might influence transportation patterns and the health of the population. Agent-based modelling has been used for some time in descriptions and studies of simulated traffic flows in road and urban networks,[104] and expansion of this approach beyond the scope of traffic engineering to incorporate aspects of health, safety, and individual behaviour offers great potential. Such an expansion could improve estimations of health and safety effects of various urban scenarios and understanding of the mechanisms that lead to optimised urban policy and planning settings.[105,106]

We contend that a modal shift that reduces reliance on private motor vehicles and an increased prevalence of walking and cycling (along with connected, accessible, and safe public transport) will lead to considerable population health benefits in relation to chronic disease. However, without the inclusion of adequate safe infrastructure, the introduction of additional cyclists and pedestrians within already highly motorised transport systems is likely to increase road trauma. Conversely, the introduction of additional cyclists and pedestrians might lead to decreased road trauma in cities with existing lower levels of motorisation (eg, Delhi) or cities with existing high levels of infrastructure that ensure that walking and cycling can be undertaken in an environment of reduced risk (eg, Copenhagen). Policies that support walking and cycling within a safe urban environment are paramount to achieve gains in overall population health. Moreover, achievement of these outcomes will require a comprehensive integrated approach, as the first paper in this Series has highlighted.[9]

Cars, cities, and health: new urban mobility

New urban mobility in which transport policies encourage walking, cycling, and public transport and reduce subsidies for private motor vehicle use are being supported by cities across many high-income countries.[107] Cities such as Helsinki (Finland) and Zurich (Switzerland) have seen substantial modal shifts from private motor vehicle use to walking, cycling, and public transport. For example, 52% of Zurich's daily VKTs are now undertaken by either walking or cycling, 19% are undertaken on public transport, and only 29% are undertaken using a private motor vehicle. The city has achieved this by limiting car parking, prioritising trams on road space, and deliberately creating congestion, with traffic signals providing access to streets to only a few cars at a time.[86]

Cities embracing new urban mobility are setting ambitious targets to achieve safe and sustainable transport over the coming years and are building infrastructure to the quality previously built for motor vehicles. For example, Helsinki (which is not typical of other European cities because it was designed during a period of considerable motorisation) has embraced sustainable mobility and is transforming its car-dependent suburbs into denser and more walkable mixed-use precincts that are linked to the city centre by rapid and frequent public transport.[108]

Extensive infrastructure is being delivered for cycling in Helsinki to support its proactive transport policy of achieving a cycling mode share of 15% of VKT by 2020.[109] Cities adopting new urban mobility are doing so in the knowledge that it delivers benefits in terms of reduced overall congestion, additional opportunities for multimodal travel, and increased efficiency. This paper also highlights the considerable health gains that residents of these cities could obtain through adoption of low levels of motorised transport.

Many countries concerned by the costs associated with the mounting burden of lifestyle-related chronic disease[94] have put in place plans and public policy initiatives that encourage increased levels of physical activity.[95–97] Although the extent to which these plans are successfully disseminated, enacted, and monitored varies,[98–102] the findings reported here suggest that government policies need to actively pursue integrated urban and transport planning and design interventions—particularly those focused towards achieving more compact cities—that support and encourage modal shifts away from private motor vehicles towards new urban mobility. Such interventions are required if city planners are to positively influence the overall health and sustainability of growing cities.

Contributors

MS, JT, and RM conceptualised the paper and contributed to the study design. MW worked on the development of the compact cities model, and JW provided feedback on modelling methods and assumptions used. MS, JW, DM, GT, IR, JT, MW, THdS, XS, BG-C, and RE contributed to writing of the paper. JT and THdS did the literature searches. JW and GT provided data, and MS, THdS, and XS contributed to data collection. MS, JT, and THdS contributed to data analysis and interpretation. MS, DM, RM, THdS, BG-C, and RE critically reviewed and edited the paper.

Declaration of interests

We declare no competing interests.

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